MoS2 shell thickness control

"Which precursors control MoS2 shell thickness in hydrothermal synthesis?" *Generated by LeapSpace*

MoS_2 shell thickness in hydrothermal synthesis is controlled primarily by the *molybdenum precursor amount/concentration*, and in some systems also by the *sulfur-source concentration* and related Mo/S ratio. The clearest direct evidence shows shell thickness increasing as the MoS_2 precursor dose rises in core–shell systems, while reactant concentration balance also affects nanosheet thickness and hierarchical growth. [114] [115] [128] [129] [130]

Which precursors matter most

Precursor / control factor	Effect on MoS_2 shell thickness	Evidence
Mo precursor concentration / amount	**Most direct control knob**; higher Mo precursor loading gives a thicker MoS_2 shell or film. [114] [115] [131]	$BiVO_4$@MoS_2 shell thickness was tuned by changing MoS_2 precursor concentration, reaching about 300 nm in one optimized sample [114]. Similar thickness growth was reported in a WS_2–MoS_2 heterostructure as the MoS_2 precursor amount increased [115].
Sulfur precursor concentration	Can shift nucleation and nanosheet assembly, affecting shell build-up and sheet thickness. [128] [129] [130]	Higher reactant concentration changed the thickness of self-assembled MoS_2 nanosheets in hierarchical nanospheres [128], and sulfur concentration was identified as a key driver of nucleation and hierarchical nanoflower formation [130].
Mo/S ratio	Tunes growth density and morphology, indirectly affecting shell thickness. [129] [130]	Mo/S molar ratio changes produced different MoS_2 nanostructures and improved performance at specific ratios [129].

Bottom line

If your goal is **thickness control**, the **Mo precursor concentration/amount** is the most defensible hydrothermal precursor variable to adjust, with **sulfur precursor concentration or the Mo/S ratio** acting as secondary growth controls that influence how densely the shell assembles. [114] [115] [128] [130]

Confidence level: **High** — multiple hydrothermal studies directly report thickness tuning through Mo precursor loading, and several others show sulfur-source concentration or Mo/S ratio shaping MoS_2 growth morphology.

References

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